

STAT 4005 - Time Series

Chapter 4 - Estimation

Agenda

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- ARIMA(p, d, q) Model

$$\phi(B)(1 - B)^d(Y_t - \mu) = \theta(B)Z_t, \text{ where } Z_t \sim WN(0, \sigma^2).$$

- Unknown order: (p, d, q)
- Unknown parameters: $(\mu, \phi_1, \phi_2, \dots, \phi_p, \theta_1, \theta_2, \dots, \theta_q, \sigma^2)$

Question:

Given the order (p, d, q) , how to estimate the unknown parameters ?

Moment Estimators

- Basic idea:

Sample Moments ($\frac{1}{n} \sum_{i=1}^n Y_i^k$) can be computed from the data
Theoretical Moments ($E(Y^k)$) depends on unknown parameters
 $\Rightarrow$ Match the sample and theoretical moments to solve for the unknown parameters

- Toy example: i.i.d. Normal: $N(\mu, \sigma^2)$

	Sample	Theoretical
1 st moment	$\bar{Y}$	μ
2 nd moment	$\frac{\sum Y_i^2}{n}$	$\mu^2 + \sigma^2$

Moment estimators:

- (match 1st moment): $\hat{\mu} = \bar{Y}$
- (match 2nd moment): $\hat{\sigma}^2 = \frac{\sum Y_i^2}{n} - \bar{Y}^2$

Moment Estimator for Time Series

Theoretical Moment	Sample Moment
μ	$\bar{Y} = \frac{1}{n} \sum_{t=1}^n Y_t$
$\gamma(0)$	$C_0 = \frac{1}{n} \sum (Y_t - \bar{Y})^2$
$\gamma(k)$	$C_k = \frac{1}{n} \sum_{t=1}^{n-k} (Y_t - \bar{Y})(Y_{t+k} - \bar{Y})$
$\rho(0)$	1
$\rho(k)$	$r_k = \frac{C_k}{C_0}$

Exercise

Find the moment estimator for the AR(1) model $(1 - \phi B)(Y_t - \mu) = Z_t$, $Z_t \sim WN(0, 1)$ in terms of observations $Y_1, \dots, Y_n$. Give the value if the observations are $(1.2, 2.3, 2.1, 1.5, 0.8, 1.2)$.

Solution:

$$\begin{aligned}\hat{\mu} &= \frac{1.2+2.3+2.1+1.5+0.8+1.2}{6} = 1.5167 \\ C_0 &= \frac{1}{6}[(1.2 - 1.5167)^2 + (2.3 - 1.5167)^2 + (2.1 - 1.5167)^2 \\ &\quad + (1.5 - 1.5167)^2 + (0.8 - 1.5167)^2 + (1.2 - 1.5167)^2] \\ &= 0.2781 \\ C_1 &= \frac{1}{6}[(1.2 - 1.5167)(2.3 - 1.5167) + (2.3 - 1.5167)(2.1 - 1.5167) \\ &\quad + (2.1 - 1.5167)(1.5 - 1.5167) + (1.5 - 1.5167)(0.8 - 1.5167) \\ &\quad + (0.8 - 1.5167)(1.2 - 1.5167)] \\ &= 0.0730\end{aligned}$$

Since $\rho(1) = \phi$, we have $\hat{\phi} = r_1 = \frac{C_1}{C_0} = 0.2626$.

Exercise

For a series Y_t with length 100, we have computed $r_1 = 0.7$, $r_2 = 0.45$, $r_3 = 0.3$, $\bar{Y} = 2$, and $C_0 = 5$. Consider an AR(2) process with a constant term θ_0 ,

$$Y_t = \theta_0 + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + Z_t, \text{ where } Z_t \sim WN(0, \sigma^2).$$

Find the estimates of ϕ_1 , ϕ_2 , θ_0 , and σ^2 by the method of moment.

Solution I

From the idea of method of moments, we can construct several estimation equation from the model.

- First, we take expectation to the two-sides of the model.
- Second, we take covariance with Y_t on both sides.
- Third, we take covariance with Y_{t-1} on both sides.
- Fourth, we take covariance with Y_{t-2} on both sides.

The above four operations yield the following four equations:

- ① $\mu = \theta_0 + \phi_1\mu + \phi_2\mu,$
- ② $\gamma(0) = \phi_1\gamma(1) + \phi_2\gamma(2) + \sigma^2,$
- ③ $\gamma(1) = \phi_1\gamma(0) + \phi_2\gamma(1),$
- ④ $\gamma(2) = \phi_1\gamma(1) + \phi_2\gamma(0),$

where $\gamma(k)$ is the lag k autocovariance function.

Solution II

To use the Method of Moment, we replace $\gamma(0), \gamma(1), \gamma(2), \mu$ by $C_0, C_1, C_2, \bar{Y}$ in (1)-(4) to solve for $\theta_0, \phi_1, \phi_2, \sigma^2$. Since

$$\begin{aligned}\bar{Y} &= 2, \\ C_0 &= 5, \\ C_1 &= r_1 C_0 = 3.5, \\ C_2 &= r_2 C_0 = 2.25,\end{aligned}$$

we have the estimating equations

$$\begin{aligned}\theta_0 + 2\phi_1 + 2\phi_2 &= 2, \\ 3.5\phi_1 + 2.25\phi_2 + \sigma^2 &= 5, \\ 5\phi_1 + 3.5\phi_2 &= 3.5, \\ 3.5\phi_1 + 5\phi_2 &= 2.25.\end{aligned}$$

Solving the above equations, we have $\hat{\phi}_1 = 0.7549$, $\hat{\phi}_2 = -0.0784$, $\hat{\sigma}^2 = 2.534$ and $\hat{\theta}_0 = 0.6471$.

Moment Estimator: MA models

$$\text{MA}(q): Y_t = Z_t - \theta_1 Z_{t-1} - \dots - \theta_q Z_{t-q} \quad Z_t \sim WN(0, \sigma^2).$$

Moment Estimator (method of moment)

- Express $\rho(k)$ in terms of unknown parameters $\theta_1, \theta_2, \dots, \theta_q$
- Replace $\rho(k)$ with r_k for $k = 1, \dots, q$, we then solve for the unknown parameters.
- Example - MA(1): $Y_t = Z_t - \theta Z_{t-1}$

$$\rho(1) = -\frac{\theta}{1+\theta^2} \quad r_1 = \frac{\sum_{t=1}^{n-1} (Y_t - \bar{Y})(Y_{t+1} - \bar{Y})}{\sum_{t=1}^n (Y_t - \bar{Y})^2}$$

The estimate satisfies

$$r_1 = -\frac{\hat{\theta}}{1+\hat{\theta}^2} \Rightarrow r_1 \hat{\theta}^2 + \hat{\theta} + r_1 = 0 \Rightarrow \hat{\theta} = \frac{-1 \pm \sqrt{1 - 4r_1^2}}{2r_1}$$

Exercise

For a MA(1) model

$$Y_t = \alpha Z_t + \beta Z_{t-1}, \quad \text{for } Z_t \stackrel{\text{iid}}{\sim} N(0, 1).$$

In terms of Y_t s, find the method of moment estimator for α and β .

Solution: First, the autocovariance function of $\gamma(k)$ can be readily found as

$$\begin{aligned}\gamma(0) &= \alpha^2 + \beta^2, \\ \gamma(1) &= \alpha\beta.\end{aligned}$$

Multiply 2 or -2 to the second equation and add to the first equation, we have

$$\begin{aligned}\gamma(0) + 2\gamma(1) &= (\alpha + \beta)^2, \\ \gamma(0) - 2\gamma(1) &= (\alpha - \beta)^2.\end{aligned}$$

Exercise

The method of moment estimator can be obtained by solving

$$\begin{aligned}(\hat{\alpha} + \hat{\beta})^2 &= C_0 + 2C_1 \\&= \frac{1}{n} \sum_{t=1}^n (Y_t - \bar{Y})^2 + \frac{2}{n} \sum_{t=1}^{n-1} (Y_t - \bar{Y})(Y_{t+1} - \bar{Y}) := S_1, \\(\hat{\alpha} - \hat{\beta})^2 &= C_0 - 2C_1 \\&= \frac{1}{n} \sum_{t=1}^n (Y_t - \bar{Y})^2 - \frac{2}{n} \sum_{t=1}^{n-1} (Y_t - \bar{Y})(Y_{t+1} - \bar{Y}) := S_2.\end{aligned}$$

Note that there is no solution if $S_1 < 0$ or $S_2 < 0$. For $S_1, S_2 \geq 0$, we have the following four pairs of possible solutions:

- (1) $\hat{\alpha} = \frac{1}{2} (\sqrt{S_1} + \sqrt{S_2})$, $\hat{\beta} = \frac{1}{2} (\sqrt{S_1} - \sqrt{S_2})$,
- (2) $\hat{\alpha} = \frac{1}{2} (\sqrt{S_1} - \sqrt{S_2})$, $\hat{\beta} = \frac{1}{2} (\sqrt{S_1} + \sqrt{S_2})$,
- (3) $\hat{\alpha} = -\frac{1}{2} (\sqrt{S_1} - \sqrt{S_2})$, $\hat{\beta} = -\frac{1}{2} (\sqrt{S_1} + \sqrt{S_2})$,
- (4) $\hat{\alpha} = -\frac{1}{2} (\sqrt{S_1} + \sqrt{S_2})$, $\hat{\beta} = -\frac{1}{2} (\sqrt{S_1} - \sqrt{S_2})$.

Exercise

Find the method of moment estimates of ϕ and θ for the stationary and invertible ARMA(1,1) process

$$Y_t = \phi Y_{t-1} + Z_t - \theta Z_{t-1}, \text{ where } Z_t \sim WN(0, \sigma^2).$$

based on the first two sample autocorrelations $r_1 = 0.2$ and $r_2 = -0.1$.

Solution

For the ARMA(1,1) model, the autocorrelation function is given by

$$\begin{aligned}\rho(1) &= \frac{\gamma(1)}{\gamma(0)} = \frac{(\theta - \phi)(\phi\theta - 1)}{1 - 2\phi\theta + \theta^2}, \\ \rho(2) &= \frac{\gamma(2)}{\gamma(0)} = \frac{\phi(\theta - \phi)(\phi\theta - 1)}{1 - 2\phi\theta + \theta^2}.\end{aligned}$$

Using the method of moment, we match $(\rho(1), \rho(2))$ to (r_1, r_2) to solve for ϕ and θ . First, note that $\frac{\rho(1)}{\rho(2)} = \frac{1}{\phi}$. As $\frac{r_1}{r_2} = -2$ we get $\hat{\phi} = -0.5$. Then, solving

$$r_1 = \frac{(\theta - \hat{\phi})(\hat{\phi}\theta - 1)}{1 - 2\hat{\phi}\theta + \theta^2}$$

for θ , we get $\hat{\theta} = -1.3$ or -0.77 . Since the series is stationary and invertible, the roots of characteristic function should lie outside the unit circle. So, we take $\hat{\theta} = -0.77$.

Yule Walker Equation (Y.W.)

- Yule-Walker (Y.W.) equations is a system of equations connecting AR parameters ϕ_i s and ACVF/ACFs:
 - Multiply Y_{t-k} to $Y_t = \phi_1 Y_{t-1} + \dots + \phi_p Y_{t-p} + Z_t$ where $k = 1, 2, \dots, p$
 - Then, take expectation
 - $\gamma(k) = \phi_1 \gamma(k-1) + \dots + \phi_p \gamma(k-p)$
 $\Rightarrow \rho(k) = \phi_1 \rho(k-1) + \dots + \phi_p \rho(k-p)$
where $k = 1, 2, \dots, p$
- Given parameters ϕ_i s, we have used Yule-Walker equation to find ACVF/ACFs (Chapter 3)
- Given **sample ACFs**, we can use Yule-Walker equation to estimate ϕ_i s

Parameter Estimators in AR Model: Yule Walker

$$\begin{aligned}\rho(1) &= \phi_1\rho(0) + \cdots + \phi_p\rho(p-1) \\ \rho(2) &= \phi_1\rho(1) + \cdots + \phi_p\rho(p-2) \\ \cdots &= \cdots + \cdots + \cdots \\ \rho(p) &= \phi_1\rho(p-1) + \cdots + \phi_p\rho(0)\end{aligned}$$

Matrix Form

$$\begin{pmatrix} \rho(1) \\ \vdots \\ \rho(p) \end{pmatrix} = \begin{pmatrix} \rho(0) & \rho(1) & \cdots & \rho(p-1) \\ \rho(1) & \rho(0) & \cdots & \rho(p-2) \\ \vdots & \vdots & \ddots & \vdots \\ \rho(p-1) & \rho(p-2) & \cdots & \rho(0) \end{pmatrix} \begin{pmatrix} \phi_1 \\ \vdots \\ \phi_p \end{pmatrix}$$

Yule-Walker Estimator (estimate $\rho(k)$ by r_k)

$$\hat{\phi} = \begin{pmatrix} \hat{\phi}_1 \\ \vdots \\ \hat{\phi}_p \end{pmatrix} = \begin{pmatrix} 1 & r_1 & \cdots & r_{p-1} \\ r_1 & 1 & \cdots & r_{p-2} \\ \vdots & \vdots & \ddots & \vdots \\ r_{p-1} & r_{p-2} & \cdots & 1 \end{pmatrix}^{-1} \begin{pmatrix} r_1 \\ \vdots \\ r_p \end{pmatrix}$$

Parameter Estimators in AR Model: Yule Walker

The Yule-Walker estimator can also be expressed in terms of the sample moments C_k because

$$\begin{aligned}\begin{pmatrix} \hat{\phi}_1 \\ \vdots \\ \hat{\phi}_p \end{pmatrix} &= \begin{pmatrix} 1 & r_1 & \cdots & r_{p-1} \\ r_1 & 1 & \cdots & r_{p-2} \\ \vdots & \vdots & \ddots & \vdots \\ r_{p-1} & r_{p-2} & \cdots & 1 \end{pmatrix}^{-1} \begin{pmatrix} r_1 \\ \vdots \\ r_p \end{pmatrix} \\ &= \begin{pmatrix} 1 \\ \frac{1}{C_0} \end{pmatrix} \begin{pmatrix} C_0 & C_1 & \cdots & C_{p-1} \\ C_1 & C_0 & \cdots & C_{p-2} \\ \vdots & \vdots & \ddots & \vdots \\ C_{p-1} & C_{p-2} & \cdots & C_0 \end{pmatrix}^{-1} \begin{pmatrix} C_1 \\ \vdots \\ C_p \end{pmatrix} \\ &= \begin{pmatrix} C_0 & C_1 & \cdots & C_{p-1} \\ C_1 & C_0 & \cdots & C_{p-2} \\ \vdots & \vdots & \ddots & \vdots \\ C_{p-1} & C_{p-2} & \cdots & C_0 \end{pmatrix}^{-1} \begin{pmatrix} C_1 \\ \vdots \\ C_p \end{pmatrix}\end{aligned}$$

Example: If the time series is $(-1.4, 0.39, 0.97, 1.5, 0.59, -2.4, -2.2, -1.5, -0.42, 0.10)$, find the Yule Walker estimate of an AR(2) model.

Solution: We have $r_1 = 0.5002$, $r_2 = -0.1451$, hence

$$\begin{pmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{pmatrix} = \begin{pmatrix} 1 & r_1 \\ r_1 & 1 \end{pmatrix}^{-1} \begin{pmatrix} r_1 \\ r_2 \end{pmatrix} = \begin{pmatrix} 0.7639 \\ -0.5272 \end{pmatrix}.$$

Exercise

Given the sample autocorrelation functions $C_0 = 1$, $C_1 = 0.416$, $C_2 = .37$, use Yule-Walker equation to estimate the parameters in the AR(2) model

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + Z_t, \quad \text{for } Z_t \stackrel{\text{iid}}{\sim} N(0, \sigma^2).$$

Solution: Multiplying suitable Y_{t-k} on both sides and take expectation, we have the Yule-Walker equations

$$\gamma(1) = \phi_1 \gamma(0) + \phi_2 \gamma(1)$$

$$\gamma(2) = \phi_1 \gamma(1) + \phi_2 \gamma(0)$$

Let $\phi = (\phi_1, \phi_2)'$, the Yule-Walker estimate is

$$\hat{\phi} = \begin{pmatrix} C_0 & C_1 \\ C_1 & C_0 \end{pmatrix}^{-1} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = \begin{pmatrix} 0.317 \\ 0.238 \end{pmatrix}.$$

Parameter Estimation in AR Model: LSE

$$\text{AR}(p): Y_t = \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} + Z_t$$

- Express as a regression problem

$$Y_t = (Y_{t-1} \cdots Y_{t-p}) \begin{pmatrix} \phi_1 \\ \vdots \\ \phi_p \end{pmatrix} + Z_t = \mathbf{Y}'_{t-1} \boldsymbol{\phi} + Z_t,$$

where $\boldsymbol{\phi} = (\phi_1 \cdots \phi_p)'$ and $\mathbf{Y}_{t-1} = (Y_{t-1} \cdots Y_{t-p})'$.

- Recall in linear regression model with p predictors and n_0 observations, we have

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}.$$

Least squares estimate of $\boldsymbol{\beta}$ is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}.$$

Least squares estimate of σ^2 is

$$\hat{\sigma}^2 = \frac{1}{n_0 - p} (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}})'(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}).$$

Least Squares Estimate (LSE) for AR Model

AR(p): $Y_t = \mathbf{Y}'_{t-1}\boldsymbol{\phi} + Z_t$

Least squares estimate of $\boldsymbol{\phi}$:

$\mathbf{Y} = (Y_{p+1}, \dots, Y_n)'$, $\boldsymbol{\varepsilon} = (Z_{p+1}, \dots, Z_n)$, $\boldsymbol{\beta} = \boldsymbol{\phi}$, and

$$\begin{aligned}\mathbf{X} &= (\mathbf{Y}_p \ \mathbf{Y}_{p+1} \cdots \mathbf{Y}_{n-1})' \\ &= \begin{pmatrix} Y_p & Y_{p-1} & \cdots & Y_1 \\ Y_{p+1} & Y_p & \cdots & Y_2 \\ \vdots & & & \vdots \\ Y_{n-1} & Y_{n-2} & \cdots & Y_{n-p} \end{pmatrix} \\ \hat{\boldsymbol{\phi}} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} \\ &= \left(\sum_{t=p+1}^n \mathbf{Y}_{t-1} \mathbf{Y}'_{t-1} \right)^{-1} \left(\sum_{t=p+1}^n \mathbf{Y}_{t-1} Y_t \right)\end{aligned}$$

Example - AR(1): $Y_t = \phi_1 Y_{t-1} + Z_t$,

The observations are $\mathbf{Y}_{t-1} = Y_{t-1}$, $t = 2, \dots, n$, i.e., $n_0 = n - 1$.

$$\hat{\phi}_1 = \frac{\sum_{t=2}^n Y_{t-1} Y_t}{\sum_{t=2}^n Y_{t-1}^2} \quad \text{and} \quad \hat{\sigma}^2 = \frac{1}{n-2} \sum_{t=2}^n (Y_t - \hat{\phi}_1 Y_{t-1})^2$$

Statistical Inference in AR Model: LSE

Asymptotic distribution of the estimator :

$$\sqrt{n}(\hat{\phi} - \phi) \rightarrow N(0, \sigma^2 \Gamma_p^{-1})$$

where

$$\Gamma_p = \begin{pmatrix} \gamma(0) & \gamma(1) & \cdots & \gamma(p-1) \\ \gamma(1) & \gamma(0) & \cdots & \gamma(p-2) \\ \vdots & \vdots & \ddots & \vdots \\ \gamma(p-1) & \gamma(p-2) & \cdots & \gamma(0) \end{pmatrix}$$

Confidence intervals or hypothesis tests can be applied to make inference on the vector parameter ϕ

- $\text{Var}(\hat{\phi}_k)$ is $\frac{\sigma^2}{n}$ times k -th diagonal entry of Γ_p^{-1} .
- $\widehat{\text{Var}}(\hat{\phi}_k)$ is $\frac{\hat{\sigma}^2}{n}$ times k -th diagonal entry of $\hat{\Gamma}_p^{-1}$
(Replace $\gamma(k)$ by C_k).
- 95% C.I.: $\hat{\phi}_k \pm 2\sqrt{\widehat{\text{Var}}(\hat{\phi}_k)}$
- To test $\phi_k = 0$ at 5% sig. level, compare $\hat{\phi}_k / \sqrt{\widehat{\text{Var}}(\hat{\phi}_k)}$ with 2.

Statistical Inference in AR Model: LSE

Example: Estimate AR(2) model: $n = 200$, $Y_t = 0.3Y_{t-1} + 0.04Y_{t-2} + Z_t$; $C_0 = 1.11$, $C_1 = 0.347$, $\hat{\sigma}^2 = 1$.

$$\hat{\sigma}^2 \hat{\Gamma}_p^{-1} = \begin{pmatrix} 1.11 & 0.347 \\ 0.347 & 1.11 \end{pmatrix}^{-1} = \begin{pmatrix} 0.998 & -0.312 \\ -0.312 & 0.998 \end{pmatrix}$$

- C.I. for ϕ_2 : $[0.04 \pm 2\sqrt{0.998/n}] = [-0.101, 0.181]$. (Not significantly different from 0)
- Testing for $H_0: \phi_1 = 0$:

$$z = \frac{\hat{\phi}_1}{\sqrt{\widehat{\text{Var}}(\hat{\phi}_1)}} = \frac{0.3}{\sqrt{0.998/200}} = 4.24 > 2.$$

We should reject H_0 at 5% significance level.

Exercise

Consider an AR(1) series of length 200 with $r_1 = 0.5$. Perform a statistical test at $\alpha = 0.05$ that $H_0: \rho_1 = 0.9$ versus $H_1: \rho_1 \neq 0.9$.

Solution: For an AR(1) model,

$$X_t = \phi X_{t-1} + Z_t,$$

the lag-1 autocorrelation function is $\rho_1 = \frac{\gamma(1)}{\gamma(0)} = \phi$. So testing whether $\rho_1 = 0.9$ is equivalent to testing whether $\phi = 0.9$. Note that we have

$$r_1 = \frac{\sum_{t=1}^{n-1} (X_t - \bar{X})(X_{t+1} - \bar{X})}{\sum_{t=1}^n (X_t - \bar{X})^2} = 0.5. \text{ Since}$$

$$\frac{\sum_{t=1}^n (X_t - \bar{X})^2}{\sum_{t=1}^{n-1} (X_t - \bar{X})^2} \rightarrow 1 \text{ and } \bar{X} \rightarrow 0 \text{ as } n \rightarrow \infty,$$

r_1 is very close to the least squares estimate

$$\hat{\phi} = \frac{\sum_{t=1}^{n-1} X_t X_{t+1}}{\sum_{t=1}^{n-1} X_t^2} \approx \frac{\sum_{t=1}^{n-1} (X_t - \bar{X})(X_{t+1} - \bar{X})}{\sum_{t=1}^{n-1} (X_t - \bar{X})^2} \approx r_1.$$

Solution (Continued)

In other words, we have $\hat{\phi} \approx 0.5$.

Using the asymptotic distribution of LSE,

$$\sqrt{n}(\hat{\phi} - \phi) \xrightarrow{d} N\left(0, \frac{\sigma^2}{\gamma(0)}\right)$$

and $\gamma(0) = \frac{\sigma^2}{1-\phi^2}$ for AR(1) model, we have the test statistic

$$Z = \frac{\sqrt{n}(\hat{\phi} - \phi)}{\sqrt{\sigma^2/\gamma(0)}} = \sqrt{\frac{n}{1-\phi^2}}(\hat{\phi} - \phi) \sim N(0, 1)$$

Under $H_o : \phi = 0.9$, the test statistic

$$|Z| = \left| \sqrt{\frac{200}{1-0.9^2}}(0.5 - 0.9) \right| = 12.98 > 1.96.$$

Therefore, we reject the null hypothesis.

Exercise

Given $\sum_{t=2}^n Y_{t-1}Y_t = 328$ and $\sum_{t=1}^{n-1} Y_t^2 = 413$ and $Y_1 = -0.2$, $Y_n = 2$ with $n = 200$, for an AR(1) model

$$Y_t = \phi Y_{t-1} + Z_t, \quad \text{for } Z_t \stackrel{\text{iid}}{\sim} N(0, \sigma^2),$$

- a) Estimate the parameters ϕ and σ^2 .
- b) Find a 95% confidence interval for ϕ .

Solution

- a) First note that

$$\sum_{t=2}^n Y_t^2 = \sum_{t=1}^{n-1} Y_t^2 + Y_n^2 - Y_1^2 = 416.96.$$

Thus, the least squares estimate of the parameters ϕ and σ^2 are

$$\hat{\phi} = \frac{\sum_{t=2}^n Y_{t-1} Y_t}{\sum_{t=1}^{n-1} Y_t^2} = 0.794$$

$$\begin{aligned}\hat{\sigma}^2 &= \frac{\sum_{t=2}^n (Y_t - \hat{\phi} Y_{t-1})^2}{n-2} \\ &= \frac{1}{198} \sum_{t=2}^{200} Y_t^2 - \frac{1}{99} \hat{\phi} \sum_{t=2}^{200} Y_t Y_{t-1} + \frac{1}{198} \hat{\phi}^2 \sum_{t=1}^{199} Y_t^2 \\ &= 0.7902.\end{aligned}$$

- b) Since $\sqrt{n}(\hat{\phi} - \phi) \sim N(0, \frac{\sigma^2}{\gamma(0)})$ and $\gamma(0) = \frac{\sigma^2}{1-\phi^2}$, the confidence interval for ϕ is

$$\left(\hat{\phi} - 1.96 \sqrt{\frac{1-\hat{\phi}^2}{n}}, \hat{\phi} + 1.96 \sqrt{\frac{1-\hat{\phi}^2}{n}} \right) = (0.7097, 0.8783).$$

Exercise

Suppose we have an AR(1) model $Y_t = \theta_0 + \phi Y_{t-1} + Z_t$ with $Z_t \sim WN(0, \sigma^2)$. We regress the series $\{Y_t\}$ against the lag-1 counterpart $\{Y_{t-1}\}$, and have the following computer output:

Coefficients:

	<i>Value</i>	<i>Std.Error</i>	<i>t value</i>	<i>Pr(> t)</i>
<i>Intercept</i>	5.6421	0.8921	6.3245	0.0000
Y_{t-1}	0.6250	0.1019	6.1335	0.0000

Residual standard error: 2.862 on 78 degrees of freedom

Multiple R-Squared: 0.3090

Find the estimates of ϕ , θ_0 , and σ^2 based on the least square method.
Give an estimate of $\mu = E(Y_t)$.

Solution

From the computer outputs, we can see immediately that $\hat{\theta}_0 = 5.6421$ and $\hat{\phi} = 0.6250$. Next, the residual standard error is given by

$$\sqrt{\hat{\sigma}^2} = 2.862.$$

So $\hat{\sigma}^2 = 2.862^2 = 8.19$. Finally, from the definition of the AR(1) model, we can see that

$$\mu = \theta_0 + \phi\mu.$$

Thus, we may define the estimator

$$\hat{\mu} = \frac{\hat{\theta}_0}{1 - \hat{\phi}} = 15.0456.$$

Conditional Least Squares: Moving Average Models

Conditional Least Squares (CLS) Method

(also called Conditional Sum of Squares (CSS))

$$\text{MA}(q): Y_t = Z_t - \theta_1 Z_{t-1} - \cdots - \theta_q Z_{t-q} \quad Z_t \sim WN(0, \sigma^2).$$

Idea:

- Find the noise sequence $\{Z_t\}$ from the observation $\{Y_t\}$ (so we require the MA model to be **invertible!**)
- Minimize the sum of squares $S_*(\theta) = \sum_{t=1}^n Z_t^2$.

Example - MA(1): $Y_t = Z_t - \theta Z_{t-1}$

- **Conditional on $Z_0 = 0$** , then
 $Z_1 = Y_1, Z_2 = Y_2 + \theta Z_1, \dots, Z_n = Y_n + \theta Z_{n-1}$.
- Minimize $S_*(\theta) = \sum_{t=1}^n Z_t^2$ by numerical optimization.
- $\hat{\sigma}^2 = \frac{1}{n} S_*(\hat{\theta})$.

Conditional Least Squares: Moving Average Models

Implementation:

Note that in R, the ARMA(p, q) model is specified as

$$Y_t - \phi_1 Y_{t-1} - \cdots - \phi_p Y_{t-p} = Z_t + \theta_1 Z_{t-1} + \cdots + \theta_q Z_{t-q}.$$

Hence, the MA(1) model $Y_t = Z_t + 0.3Z_{t-1}$ should be specified as follows.

```
set.seed(4005)
y=arima.sim(2000,model=list(ma=c(0.3)))
get.z=function(theta){
  z=rep(0,length(y))
  z[1]=y[1]
  for (k in 2:length(y)){ z[k]=y[k]-theta*z[k-1]}
  return(sum(z^2))
}
optim(0.1,get.z)
```

Conditional Least Squares: ARMA Model

- **Method of moment** become very tedious on ARMA models due to the complicated expression of $\gamma(k)$.
- **Conditional least squares** can be applied similarly as in MA models.

Example: ARMA(1,1): $Y_t - \phi Y_{t-1} = Z_t - \theta Z_{t-1}$,

- Conditional on $Z_0 = 0, Y_0 = 0$,
 $Z_1(\phi, \theta) = Y_1$,
 $Z_2(\phi, \theta) = Y_2 - \phi Y_1 + \theta Z_1$,
...
 $Z_n(\phi, \theta) = Y_n - \phi Y_{n-1} + \theta Z_{n-1}$.
- Minimizes $S_*(\phi, \theta) = \sum_{t=1}^n Z_t^2(\phi, \theta)$ by numerical optimization.
- $\hat{\sigma}^2 = \frac{1}{n-1} S_*(\hat{\phi}, \hat{\theta})$.

Exercise

Consider an MA(1) process, $Y_t = Z_t - \theta Z_{t-1}$. Based on a series of length 4, we observed $Y_1 = 0$, $Y_2 = 0$, $Y_3 = 2$, and $Y_4 = 1$.

- a) Find the conditional least-square estimate of θ .
- b) Find an estimate of σ^2 .

Solution: (a) Condition on $Z_0 = 0$, for a given θ we estimate the white noise Z_t by

- $Z_1 = Y_1$,
- $Z_2 = Y_2 + \theta Z_1 = Y_2 + \theta Y_1$,
- $Z_3 = Y_3 + \theta Z_2 = Y_3 + \theta Y_2 + \theta^2 Y_1$,
- $Z_4 = Y_4 + \theta Z_3 = Y_4 + \theta Y_3 + \theta^2 Y_2 + \theta^3 Y_1$.

The conditional least squares criterion is

$$\begin{aligned}\sum_{t=1}^4 Z_t^2 &= Y_1^2 + (Y_2 + \theta Y_1)^2 + (Y_3 + \theta Y_2 + \theta^2 Y_1)^2 \\ &\quad + (Y_4 + \theta Y_3 + \theta^2 Y_2 + \theta^3 Y_1)^2 \\ &= 2^2 + (1 + 2\theta)^2 \\ &= 4\theta^2 + 4\theta + 5.\end{aligned}$$

The conditional least squares estimator of θ is computed by minimizing $\sum_{t=1}^4 Z_t^2$. By differentiating $\sum_{t=1}^4 Z_t^2$ with respect to θ , $\hat{\theta}$ is obtained by solving $8\theta + 4 = 0$. Thus we have $\hat{\theta} = -\frac{1}{2}$.

(b) With the estimate $\hat{\theta} = -\frac{1}{2}$, the white noise Z_t can be estimated by

$$\hat{Z}_1 = Y_1 = 0$$

$$\hat{Z}_2 = Y_2 + \hat{\theta}Y_1 = 0$$

$$\hat{Z}_3 = Y_3 + \hat{\theta}Y_2 + \hat{\theta}^2Y_1 = 2$$

$$\hat{Z}_4 = Y_4 + \hat{\theta}Y_3 + \hat{\theta}^2Y_2 + \hat{\theta}^3Y_1 = 0$$

The conditional least squares estimate for the variance σ_a^2 can be obtained by

$$\hat{\sigma}^2 = \frac{1}{4} \sum_{t=1}^4 \hat{Z}_t^2 = 1.$$

Exercice

For an ARMA(1,1) model

$$Y_t = \phi Y_{t-1} + Z_t + \theta Z_{t-1}, \quad \text{for } Z_t \stackrel{\text{iid}}{\sim} N(0, \sigma^2),$$

write down $\sum_{t=1}^3 Z_t^2$ in terms of ϕ and θ and $\{Y_1, Y_2, Y_3\}$.

Solution: Condition on $Z_0 = 0$ and $Y_0 = 0$, we have

$$Z_1 = Y_1,$$

$$Z_2 = Y_2 - \phi Y_1 - \theta Z_1 = Y_2 - (\phi + \theta)Y_1,$$

$$Z_3 = Y_3 - \phi Y_2 - \theta Z_2 = Y_3 - (\phi + \theta)Y_2 + \theta(\phi + \theta)Y_1.$$

Thus

$$\begin{aligned} \sum_{t=1}^3 Z_t^2 &= (1 + (\phi + \theta)^2 + \theta^2(\phi + \theta)^2)Y_1^2 + (1 + (\phi + \theta)^2)Y_2^2 + Y_3^2 \\ &\quad - 2((\phi + \theta) + \theta(\phi + \theta)^2)Y_1Y_2 + 2\theta(\phi + \theta)Y_1Y_3 - 2(\phi + \theta)Y_2Y_3. \end{aligned}$$

Maximum Likelihood Estimator (MLE)

- MLE = Most probable parameter value of the statistical model inferred from the observed data
- Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with probability density function $f(x, \theta)$
 - $f(X_i, \theta)$ is the probability of observing X_i
 - The likelihood function $L(\mathbf{X}, \theta) = \prod_{i=1}^n f(X_i, \theta)$ is the probability of observing the data set, where $\mathbf{X} = (X_1, \dots, X_n)$.
- The MLE $\hat{\theta}$ maximizes $L(\mathbf{X}, \theta)$
 - The most probable parameter value such that the current data set is obtained

Maximum Likelihood Estimator

MLE for Time Series

- **Main Idea:** Write down the **joint p.d.f** of $Y_1, Y_2, \dots, Y_n$.
- Example: AR(1): $Y_t = \phi Y_{t-1} + Z_t$, $Z_t \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$.
Conditional on Y_1 , $(Y_2, \dots, Y_n)$ and $(Z_2, \dots, Z_n)$ are of one-to-one correspondence

$$\begin{aligned} f(Y_2, \dots, Y_n | Y_1) &= f(Z_2, \dots, Z_n | Y_1) \\ &= \left(\frac{1}{2\pi\sigma^2} \right)^{\frac{n-1}{2}} \exp \left(-\frac{1}{2\sigma^2} \sum_{t=2}^n Z_t^2 \right) \\ &= \left(\frac{1}{2\pi\sigma^2} \right)^{\frac{n-1}{2}} \exp \left(-\frac{1}{2\sigma^2} \sum_{t=2}^n (Y_t - \phi Y_{t-1})^2 \right) \end{aligned}$$

Maximum Likelihood Estimator

Stationarity implies that $Y_1 \sim N(0, \frac{\sigma^2}{1-\phi^2})$, the likelihood function is

$$\begin{aligned} & L(Y_1, Y_2, \dots, Y_n) \\ = & f(Y_2, \dots, Y_n | Y_1) f(Y_1) \\ = & f(Y_2, \dots, Y_n | Y_1) \frac{\sqrt{1-\phi^2}}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(1-\phi^2)Y_1^2\right) \\ = & \sqrt{1-\phi^2} \left(\frac{1}{2\pi\sigma^2}\right)^{\frac{n}{2}} \exp\left\{-\frac{1}{2\sigma^2} \left[\sum_{t=2}^n (Y_t - \phi Y_{t-1})^2 + (1-\phi^2)Y_1^2\right]\right\}. \end{aligned}$$

The parameter $(\hat{\phi}, \hat{\sigma})$ could then be obtained by maximizing $L(Y_1, \dots, Y_n)$, or equivalently $\log L(Y_1, \dots, Y_n)$, with respect to (ϕ, σ) by numerical methods

Maximum Likelihood Estimator: Two standard approaches

Approach 1: Iterative Conditioning

- Decompose the joint distribution into a series of conditional distributions.

$$\begin{aligned} & f(Y_1, \dots, Y_n) \\ = & \frac{f(Y_1, \dots, Y_n)}{f(Y_1, \dots, Y_{n-1})} \frac{f(Y_1, \dots, Y_{n-1})}{f(Y_1, \dots, Y_{n-2})} \dots \frac{f(Y_1, Y_2)}{f(Y_1)} f(Y_1) \\ = & \left\{ \prod_{t=2}^n f(Y_t | Y_{t-1}, \dots, Y_1) \right\} f(Y_1) \end{aligned}$$

- $f(Y_t | Y_{t-1}, \dots, Y_1)$ may be easy to write down:
e.g. AR(p) model:

$$Y_t = \phi_1 Y_{t-1} + \dots + \phi_p Y_{t-p} + Z_t, \quad Z_t \sim N(0, \sigma^2)$$

$$Y_t | Y_{t-1}, \dots, Y_1 \sim N(\phi_1 Y_{t-1} + \dots + \phi_p Y_{t-p}, \sigma^2)$$

Maximum Likelihood Estimator: Two standard approaches

Approach 2: Multivariate Normal Distribution

- In **causal** ARMA models, if $\{Z_t\}$ are normally distributed, then $\{Y_t\}$ are also normal (as $Y_t = \sum_{j=1}^{\infty} \psi_j Z_{t-j}$ is a sum of normal noises)
- $\{Y_1, \dots, Y_n\}$ follows multivariate normal distribution, which is characterized by the ACVFs:

$$f_{Y_1, \dots, Y_n}(y_1, \dots, y_n) = \frac{1}{(2\pi)^{\frac{n}{2}} |\Sigma|^{\frac{1}{2}}} e^{-\frac{1}{2} \mathbf{y}' \Sigma^{-1} \mathbf{y}},$$

where $\mathbf{y} = (y_1, \dots, y_n)'$ and

$$\Sigma = (\gamma(|i - j|)) = \begin{pmatrix} \gamma(0) & \gamma(1) & \dots & \gamma(n-1) \\ \gamma(1) & \gamma(0) & \dots & \gamma(n-2) \\ & & \ddots & \\ \gamma(n-1) & \gamma(n-2) & \dots & \gamma(0) \end{pmatrix}$$

Exercise

For MA(1) model

$$Y_t = Z_t + \theta Z_{t-1}, \quad \text{for } Z_t \stackrel{\text{iid}}{\sim} N(0, \sigma^2).$$

write down the likelihood function for the observations $\{Y_1, Y_2, Y_3\}$.

Solution: Note that the ACVF function is $\gamma(0) = \sigma^2(1 + \theta^2)$, $\gamma(1) = \sigma^2\theta$ and $\gamma(k) = 0$ for $k \geq 2$. Thus the likelihood function is

$$l(\theta, \sigma^2) = f(y_1, y_2, y_3) = \frac{1}{(2\pi)^{3/2} |\Sigma|^{1/2}} e^{-\frac{1}{2} \mathbf{y}' \Sigma^{-1} \mathbf{y}},$$

where $\mathbf{y} = (y_1, y_2, y_3)'$ and

$$\Sigma = \sigma^2 \begin{pmatrix} 1 + \theta^2 & \theta & 0 \\ \theta & 1 + \theta^2 & \theta \\ 0 & \theta & 1 + \theta^2 \end{pmatrix}.$$

Maximum Likelihood Estimation using R

- ARMA(p, q) model:

$$Y_t = \mu + \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} + Z_t + \theta_1 Z_{t-1} + \cdots + \theta_q Z_{t-q}, \\ Z_t \sim WN(0, \sigma^2)$$

- Except for some simple models such as AR(1), we have to employ numerical optimization for the estimation.
- `arima(x, order = c(0L, 0L, 0L),
seasonal = list(order = c(0L, 0L, 0L), period = NA),
xreg = NULL, include.mean = TRUE,
fixed = NULL, init = NULL,
method = c("CSS-ML", "ML", "CSS"))`
- Examples:
 - `set.seed(4005)`
 - `x=arima.sim(n=1000,
model=list(ar=c(0.7,-0.12),ma=c(0.7)))`
 - `arima(x,order=c(2,0,1))`
 - `arima(x,order=c(2,0,1),include.mean=F)`

Partial ACF (PACF)

- PACF measures the correlation between Y_{k+1} and Y_1 that is not explained by $Y_k, \dots, Y_2$.
- Formally, ϕ_{kj} , $j = 1, \dots, k$ are the coefficients in the representation

$$Y_{k+1} = \phi_{k1}Y_k + \phi_{k2}Y_{k-1} + \dots + \phi_{k,k-1}Y_2 + \phi_{kk}Y_1 + Z_{k+1}. \quad (1)$$

The PACF of lag order k is defined to be ϕ_{kk} . Note that this is not a representation of the "true model".

- On both sides of (1), multiply by Y_{k+1-j} , for $j = 1, 2, \dots, k$, take expectation, and then divide by $\gamma(0)$, we obtain

$$\rho(j) = \phi_{k1}\rho(|j-1|) + \phi_{k2}\rho(|j-2|) + \dots + \phi_{kk}\rho(|j-k|),$$

for $j = 1, 2, \dots, k$.

- Hence, $\phi_{k1}, \dots, \phi_{kk}$ can be obtained by solving

$$\begin{pmatrix} \rho(1) \\ \vdots \\ \rho(k) \end{pmatrix} = \begin{pmatrix} \rho(0) & \cdots & \rho(k-1) \\ \vdots & \ddots & \vdots \\ \rho(k-1) & \cdots & \rho(0) \end{pmatrix} \begin{pmatrix} \phi_{k1} \\ \vdots \\ \phi_{kk} \end{pmatrix}.$$

Partial ACF (PACF)

Consider the optimization problem

$$\min_{\beta_1, \dots, \beta_k} E[(Y_{k+1} - \beta_1 Y_k - \dots - \beta_k Y_1)^2]$$

Differentiation with respect to β_j , $j = 1, \dots, k$, we have

$$\begin{aligned} & \frac{d}{d\beta_j} E[(Y_{k+1} - \beta_1 Y_k - \dots - \beta_k Y_1)^2] \\ &= E\left[\frac{d}{d\beta_j} (Y_{k+1} - \beta_1 Y_k - \dots - \beta_k Y_1)^2\right] \\ &= E[2(Y_{k+1} - \beta_1 Y_k - \dots - \beta_k Y_1)(-Y_{k+1-j})] \\ &= -2\gamma(0)(\rho(j) - \beta_1 \rho(|j-1|) - \beta_2 \rho(|j-2|) - \dots - \beta_k \rho(|j-k|)). \end{aligned}$$

Setting the derivatives to zero for $j = 1, \dots, k$, and rename the optimal solution as $\beta_j^* = \phi_{kj}$, we obtain again

$$\begin{pmatrix} \rho(1) \\ \vdots \\ \rho(k) \end{pmatrix} = \begin{pmatrix} \rho(0) & \cdots & \rho(k-1) \\ \vdots & \ddots & \vdots \\ \rho(k-1) & \cdots & \rho(0) \end{pmatrix} \begin{pmatrix} \phi_{k1} \\ \vdots \\ \phi_{kk} \end{pmatrix}.$$

If the time series follows AR(p) model:

$$Y_t = \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} + Z_t$$

- $\phi_{11}, \phi_{22}, \dots, \phi_{pp}$ are non-zero, i.e. Y_t is linearly related to Y_{t-k} for $k \leq p$
- $Y_k - \phi_1 Y_{k-1} - \cdots - \phi_p Y_{k-p} = Z_k$ is uncorrelated with all $\{Y_j\}_{j < k-p}$, thus, **additional** $Y_{k-p-1}, Y_{k-p-2}, \dots, Y_1$ **cannot explain** Y_k **in a linear relation**, so

$$\phi_{kk} = 0 \quad \text{for all } k > p$$

Computational formula for PACF

$$\begin{pmatrix} \phi_{k1} \\ \vdots \\ \phi_{kk} \end{pmatrix} = \begin{pmatrix} \rho(0) & \cdots & \rho(k-1) \\ \vdots & \ddots & \vdots \\ \rho(k-1) & \cdots & \rho(0) \end{pmatrix}^{-1} \begin{pmatrix} \rho(1) \\ \vdots \\ \rho(k) \end{pmatrix}$$

- $k = 1, \phi_{11} = \rho(1).$

- $k = 2, \phi_{22} = \frac{\begin{vmatrix} 1 & \rho(1) \\ \rho(1) & \rho(2) \end{vmatrix}}{\begin{vmatrix} 1 & \rho(1) \\ \rho(1) & 1 \end{vmatrix}} = \frac{\rho(2) - \rho(1)^2}{1 - \rho(1)^2}.$

- $k = 3, \phi_{33} = \frac{\begin{vmatrix} 1 & \rho(1) & \rho(2) \\ \rho(1) & 1 & \rho(3) \\ \rho(2) & \rho(1) & \rho(3) \end{vmatrix}}{\begin{vmatrix} 1 & \rho(1) & \rho(2) \\ \rho(1) & 1 & \rho(1) \\ \rho(2) & \rho(1) & 1 \end{vmatrix}} = \frac{\rho(1)^3 - \rho(1)^2 \rho(3) + \rho(1) \rho(2)^2 - 2\rho(1) \rho(2) + \rho(3)}{2\rho(1)^2 \rho(2) - 2\rho(1)^2 - \rho(2)^2 + 1}.$

Computational formula for PACF

$$\begin{pmatrix} \phi_{k1} \\ \vdots \\ \phi_{kk} \end{pmatrix} = \begin{pmatrix} \rho(0) & \cdots & \rho(k-1) \\ \vdots & \ddots & \vdots \\ \rho(k-1) & \cdots & \rho(0) \end{pmatrix}^{-1} \begin{pmatrix} \rho(1) \\ \vdots \\ \rho(k) \end{pmatrix}$$

Example: AR(1): $Y_t = \phi Y_{t-1} + Z_t$. Recall that $\rho(k) = \phi^{|k|}$.

1st lag PACF is:

$$\phi_{11} = \rho(0)^{-1} \rho(1) = \rho(1) = \phi$$

For 2nd lag PACF,

$$\begin{aligned} \begin{pmatrix} \phi_{21} \\ \phi_{22} \end{pmatrix} &= \begin{pmatrix} \rho(0) & \rho(1) \\ \rho(1) & \rho(0) \end{pmatrix}^{-1} \begin{pmatrix} \rho(1) \\ \rho(2) \end{pmatrix} \\ &= \frac{1}{1 - \phi^2} \begin{pmatrix} 1 & -\phi \\ -\phi & 1 \end{pmatrix} \begin{pmatrix} \phi \\ \phi^2 \end{pmatrix} = \begin{pmatrix} \phi \\ 0 \end{pmatrix} \end{aligned}$$

The 2nd lag PACF is $\phi_{22} = 0$.

Computational formula for PACF

Example: MA(1): $Y_t = Z_t + \theta Z_{t-1}$. Recall $\rho(1) = \frac{\theta}{1+\theta^2}$, $\rho(k) = 0$ ($k \geq 2$).
1st lag PACF is:

$$\phi_{11} = \rho(0)^{-1} \rho(1) = \rho(1) = \frac{\theta}{1 + \theta^2}$$

For 2nd lag PACF,

$$\begin{aligned} \begin{pmatrix} \phi_{21} \\ \phi_{22} \end{pmatrix} &= \begin{pmatrix} 1 & \rho(1) \\ \rho(1) & 1 \end{pmatrix}^{-1} \begin{pmatrix} \rho(1) \\ 0 \end{pmatrix} \\ &= \frac{1}{1 - \rho(1)^2} \begin{pmatrix} 1 & -\rho(1) \\ -\rho(1) & 1 \end{pmatrix} \begin{pmatrix} \rho(1) \\ 0 \end{pmatrix} = \frac{1}{1 - \rho(1)^2} \begin{pmatrix} \rho(1) \\ -\rho(1)^2 \end{pmatrix} \end{aligned}$$

The 2nd lag PACF is

$$\phi_{22} = \frac{-\rho(1)^2}{1 - \rho(1)^2} = -\frac{\theta^2}{1 + \theta^2 + \theta^4}.$$

Computational formula for PACF

For k th lag PACF, we solve

$$\begin{pmatrix} 1 & \rho(1) & & & \\ \rho(1) & 1 & \rho(1) & & \\ & \rho(1) & 1 & \rho(1) & \\ & \dots & & \rho(1) & 1 & \rho(1) \\ & & & \rho(1) & 1 \end{pmatrix} \begin{pmatrix} \phi_{k1} \\ \phi_{k2} \\ \vdots \\ \phi_{kk} \end{pmatrix} = \begin{pmatrix} \rho(1) \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

Advanced techniques in difference equations and tedious algebras show that

$$\phi_{kk} = -\frac{(-\theta)^k(1 - \theta^2)}{1 - \theta^{2(k+1)}}.$$

We see that this PACF is roughly exponential decaying.

Exercise

For a MA(2) model

$$Y_t = Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2}, \quad \text{for } Z_t \stackrel{\text{iid}}{\sim} N(0, 1)$$

find the lag-2 PACF for Y_t .

Solution: Lag-2 PACF is the β_2 that achieve the minimum of

$$E(Y_t - \beta_1 Y_{t-1} - \beta_2 Y_{t-2})^2.$$

Differentiating the above expectation with respect to β_1 and β_2 , the best (β_1, β_2) is the (ϕ_{21}, ϕ_{22}) that satisfies

$$\gamma(1) - \phi_{21}\gamma(0) - \phi_{22}\gamma(1) = 0,$$

$$\gamma(2) - \phi_{21}\gamma(1) - \phi_{22}\gamma(0) = 0,$$

or

$$\begin{pmatrix} \gamma(0) & \gamma(1) \\ \gamma(1) & \gamma(0) \end{pmatrix} \begin{pmatrix} \phi_{21} \\ \phi_{22} \end{pmatrix} = \begin{pmatrix} \gamma(1) \\ \gamma(2) \end{pmatrix}.$$

Solution

Solving the above equations, the lag-2 PACF ϕ_{22} is

$$\phi_{22} = \frac{\gamma(0)\gamma(2) - \gamma(1)^2}{\gamma(0)^2 - \gamma(1)^2}. \quad (2)$$

For an MA(2) model with $\sigma^2 = 1$, the autocovariance functions is

$$\begin{aligned}\gamma(0) &= 1 + \theta_1^2 + \theta_2^2, \\ \gamma(1) &= \theta_1(1 + \theta_2), \\ \gamma(2) &= \theta_2, \\ \gamma(k) &= 0 \quad \text{for } |k| \geq 3.\end{aligned}$$

Substitute the autocovariance functions into (2) yields

$$\phi_{22} = \frac{\theta_2(1 + \theta_1^2 + \theta_2^2) - \theta_1^2(1 + \theta_2)^2}{(1 + \theta_1^2 + \theta_2^2)^2 - \theta_1^2(1 + \theta_2)^2}.$$

Sample PACF

$$\begin{pmatrix} \hat{\phi}_{k1} \\ \vdots \\ \hat{\phi}_{kk} \end{pmatrix} = \begin{pmatrix} r_0 & \cdots & r_{k-1} \\ \vdots & \ddots & \vdots \\ r_{k-1} & \cdots & r_0 \end{pmatrix}^{-1} \begin{pmatrix} r_1 \\ \vdots \\ r_k \end{pmatrix}$$

Example:

Data = $(-0.63, -1.8, -0.98, -0.67, -1.14, -1.67, -2.35, -1.70)$.

- `x=c(-0.63, -1.8, -0.98, -0.67, -1.14, -1.67, -2.35, -1.70)`
- `r=acf(x)$ acf[1:1];solve(toeplitz(r),acf(x)$ acf[2:2])`
lag 1 pacf
- `r=acf(x)$ acf[1:2];solve(toeplitz(r),acf(x)$ acf[2:3])`
lag 2 pacf
- `r=acf(x)$ acf[1:3];solve(toeplitz(r),acf(x)$ acf[2:4])`
lag 3 pacf
- `pacf(x)$ acf` # all pacf

Theorem of PACF for AR Models

For an $AR(p)$ Model,

$$\sqrt{n}\hat{\phi}_{kk} \sim N(0, 1) \text{ for } k > p$$

To test whether ϕ_{kk} is significant,

- **Significant** if $|\hat{\phi}_{kk}| > \frac{2}{\sqrt{n}}$
 $\Rightarrow$ should model the time series by $AR(p)$, $p \geq k$
- **NOT Significant** otherwise
 $\Rightarrow$ should model the time series by $AR(p)$, $p < k$

Example:

- Data = $(-0.63, -1.8, -0.98, -0.67, -1.14, -1.67, -2.35, -1.70)$.
- Are the PACFs significant?

Using ACF and PACF to determine the order of AR and MA model

AR(p) Model -

- ACF plot: No clear pattern except AR(1) shows exponential decay pattern
- PACF plot: Cut-off at lag p

MA(q) Model -

- ACF plot: Cut-off at lag q
- PACF plot: No clear pattern except MA(1) shows exponential decay pattern

Remark: No clear pattern in ACF/PACF plots for ARMA(p, q) models.

Examples on PACF

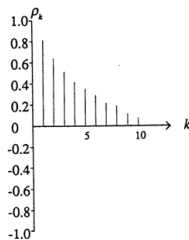
Try:

- `x=arima.sim(1000,model=list(ar=c(0.2,0.4,0.3)))`
- `pacf(x)`
- Which model will you decide?

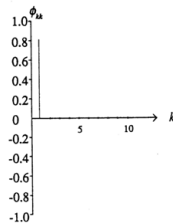
Examples on ACF and PACF

AR(1)

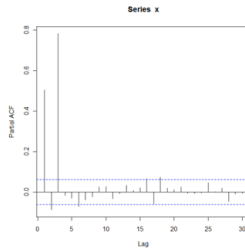
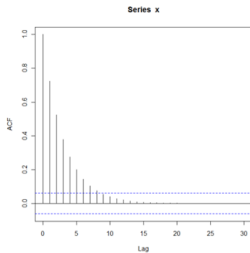
ACF



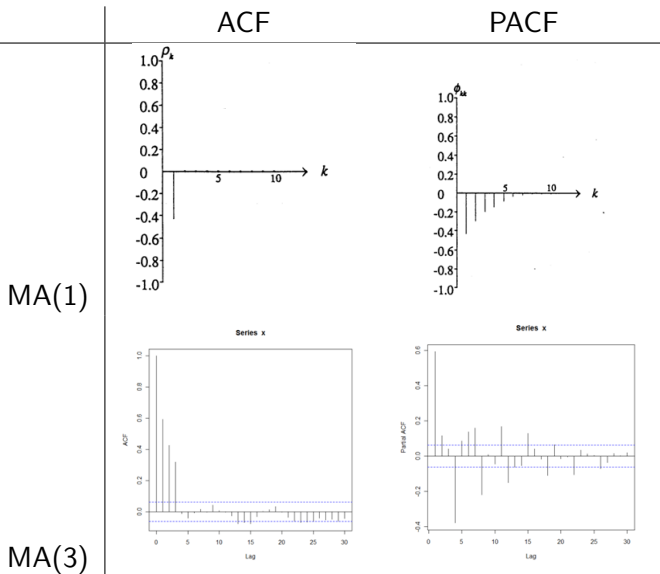
PACF



AR(3)



Examples on ACF and PACF



Exercise

From a series of 100 observations, we calculate $r_1 = -0.5$, $r_2 = 0.3$, $r_3 = -0.2$, $r_4 = 0.1$, $|r_k| < 0.09$ for $k > 4$. On the basis of this information alone, what ARIMA model would we tentatively specify for the series?

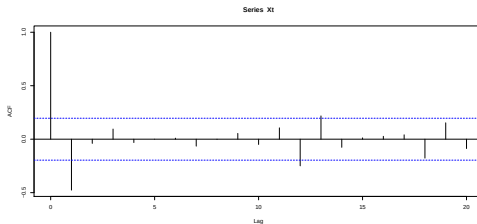
Solution:

- The ACVF decays to zero rapidly, so the series is stationary and should not have "integrated" part.
- Since the sample estimator of autocorrelation function alternates between negative and positive signs, an AR(1) model with negative coefficient may be tentatively specified.

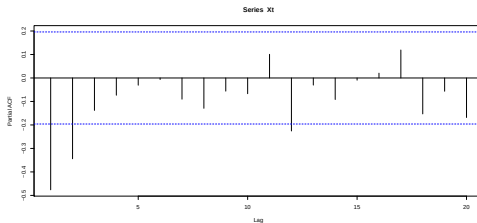
Exercise

The following figures display the ACF and PACF of the process $\{X_t\}$, respectively. What model would you suggest for $\{X_t\}$?

ACF



PACF



Solution

The ACF plot has a cutoff at lag one, which suggests an MA(1) model. Also, the PACF is exponentially decaying, which further supports the MA(1) model.

Nevertheless, the ACF and PACF at lag-12 are marginally significant, suggesting that there could be a seasonal effect.

Therefore, we may study models such as

- MA(1): $Y_t = (1 - \theta B)Z_t$,
- SARIMA(0, 0, 1) $\times$ (1, 0, 0)₁₂: $(1 - \Phi_s B^{12})Y_t = (1 - \theta B)Z_t$,
- SARIMA(0, 0, 1) $\times$ (0, 0, 1)₁₂: $Y_t = (1 - \Theta_s B^{12})(1 - \theta B)Z_t$,

where $Z_t \sim WN(0, \sigma^2)$.

- ACF and PACF are graphical methods to determine the order of AR and MA model
- It is more desirable to have a systematic order selection criterion for a general ARMA model
- Some common model selection criterion:
 - FPE (Final Prediction Error)
 - AIC (Akaike's Information Criterion)
 - BIC (Bayesian Information Criterion)

Final Prediction Error (FPE)

- $FPE = \hat{\sigma}^2 \left(\frac{n+p}{n-p} \right)$ where $\hat{\sigma}^2$ is the MLE of σ^2 , n is the sample size, p is the order of the AR component.

- **Idea of FPE:**

- For an $AR(p)$ Model, the Mean Square Error of parameter estimate is

$$MSE = E(\hat{\phi} - \phi)'(\hat{\phi} - \phi) \approx \sigma^2 \left(\frac{n+p}{n} \right) \quad (3)$$

- The best unbiased estimator for σ^2 is $\hat{\sigma}^2 \frac{n}{n-p}$
 - Substituting σ^2 by $\hat{\sigma}^2 \frac{n}{n-p}$ in (3) gives the best estimator of MSE (define as **FPE**).
 - We find the model ($AR(1)$, $AR(2)$, $AR(3)$, etc) that minimizes FPE.
 - Note the trade-off between goodness of fit and the model complexity

Order Selection - Akaike's Information Criterion

Akaike's Information Criterion (AIC)

AIC

$$-2 \log L_{\mathbf{X}} \left(\hat{\beta}, \frac{S_x(\hat{\beta})}{n} \right) + 2(p + q + 1)$$

AICC (AIC corrected)

$$-2 \log L_{\mathbf{X}} \left(\hat{\beta}, \frac{S_x(\hat{\beta})}{n} \right) + \frac{2(p + q + 1)n}{n - p - q - 2}$$

where

$$S_x(\hat{\beta}) = \sum_{t=1}^n (X_t - \hat{\phi}_1 X_{t-1} - \cdots - \hat{\phi}_p X_{t-p} - \hat{\theta}_1 Z_{t-1} - \cdots - \hat{\theta}_q Z_{t-q})^2$$

$$\hat{\beta} = (\hat{\phi}_1, \dots, \hat{\phi}_p, \hat{\theta}_1, \dots, \hat{\theta}_q) \quad \text{is the MLE}$$

$$L_{\mathbf{X}}(\hat{\beta}, \hat{\sigma}^2) = \left(\frac{1}{2\pi\hat{\sigma}^2} \right)^{\frac{n}{2}} \exp \left[-\frac{1}{2\hat{\sigma}^2} S_x(\hat{\beta}) \right]$$

Order Selection - Akaike's Information Criterion

Idea -

- AIC and AICC estimates the expected log-likelihood function
- $E[\log L_Y(\hat{\beta}, \hat{\sigma}^2)]$ where
 - $\hat{\beta}, \hat{\sigma}^2$ are MLE from the observation $\mathbf{X} = \{X_1, \dots, X_n\}$
 - Y is a 'new' observation independent of $\mathbf{X}$
- $\mathbf{X}$ and Y are different to avoid the problem of overfitting
 - $\hat{\beta}, \hat{\sigma}^2$ were chosen to maximize $\log L_{\mathbf{X}}(\beta, \sigma^2)$
- We find the model (AR(1), ARMA(1,1), etc) that minimizes AIC.
- Note the trade-off between model complexity and goodness of fit

Order Selection - Bayesian Information Criterion

Bayesian Information Criterion (BIC)

$$BIC = (n - p - q) \log \left[\frac{n\hat{\sigma}^2}{n - p - 1} \right] + n(1 + \log \sqrt{2\pi}) \\ + (p + q) \log \left[\frac{\sum_{i=1}^n X_i^2 - n\hat{\sigma}^2}{p + q} \right]$$

- Motivated by Bayesian argument:
 - BIC is linearly dependent on the $-\log$ of posterior probability of a particular model given the data
- We find the model (AR(1), ARMA(1,1), etc) that minimizes BIC.
- Consistent order selection procedure
 - As $n \rightarrow \infty$, the order selected by BIC will be equal to the true order with probability going to 1
 - AIC/FPE are not consistent (i.e. could get a wrong model even for large sample), but achieve minimum prediction error

Example

- $FPE = \hat{\sigma}^2 \left(\frac{n+p}{n-p} \right)$
- $AIC = -2 \log L \left(\hat{\beta}, \frac{S_x(\hat{\beta})}{n} \right) + 2(p+q+1)$
- $BIC = (n-p-q) \log \left[\frac{n\hat{\sigma}^2}{n-p-1} \right] + n(1 + \log \sqrt{2\pi}) + (p+q) \log \left[\frac{\sum_{i=1}^n X_i^2 - n\hat{\sigma}^2}{p+q} \right]$

```
IC=function(x,order.input=c(1,0,1)){  
  fit=arima(x,order=order.input,include.mean=F);  
  n=length(x);p=order.input[1];q=order.input[3];sig=fit$ sigma2  
  FPE=sig*(n+p)/(n-p);AIC=fit$ aic  
  BIC=(n-p-q)*log(n*sig/(n-p-1))+n*(1+log(sqrt(2*pi)))+  
  (p+q)*log((sum(x^2)-n*sig)/(p+q)); out=list(FPE,AIC,BIC);  
  names(out)=c("FPE","AIC","BIC"); return(out) }
```

Example:

- Data = (-0.63, -1.8, -0.98, -0.67, -1.14, -1.67, -2.35, -1.70).
- Is ARMA(1,1) or AR(2) better?
 - $x=c(-0.63, -1.8, -0.98, -0.67, -1.14, -1.67, -2.35, -1.70)$
 - $x=x-\text{mean}(x)$; $IC(x, \text{order}=c(1,0,1))$; $IC(x, \text{order}=c(2,0,0))$

Exercise

A number of models are fitted to the data set with $n = 400$ and the following results are obtained:

- a) ARMA(1,2) log-likelihood = -634
- b) ARMA(2,3) log-likelihood = -636
- c) ARMA(4,3) log-likelihood = -641
- d) ARMA(1,1) log-likelihood = -630

Which model should you choose in terms of AICC?

Solution: By the AICC formula $-2 \log L_{\mathbf{X}} + \frac{2n(p+q+1)}{n-p-q-2}$, the AICC value for the four models are computed respectively as

- a) AICC = 1276
- b) AICC = 1284
- c) AICC = 1298
- d) AICC = 1266

So we choose model ARMA(1,1) which has the smallest value of AICC.

Residual Analysis

Fitted values and residuals:

- Once we obtained the estimates, say $\hat{\phi}$ and $\hat{\theta}$ in ARMA(1,1) model $Y_t = \phi Y_{t-1} + Z_t + \theta Z_{t-1}$, we can compute the *fitted values*, $\{\hat{Y}_t\}$, and *residuals*, $\{\hat{Z}_t\}$, by the following recursions:

①	$\hat{Y}_1 = 0;$	$\hat{Z}_1 = Y_1 - \hat{Y}_1$
②	$\hat{Y}_2 = \hat{\phi}Y_1 + \hat{\theta}\hat{Z}_1;$	$\hat{Z}_2 = Y_2 - \hat{Y}_2$
③	$\hat{Y}_3 = \hat{\phi}Y_2 + \hat{\theta}\hat{Z}_2;$	$\hat{Z}_3 = Y_3 - \hat{Y}_3$
④		
⑤	$\hat{Y}_n = \hat{\phi}Y_{n-1} + \hat{\theta}\hat{Z}_{n-1};$	$\hat{Z}_n = Y_n - \hat{Y}_n$

- If the model fits the data well, then the fitted values $\{\hat{Y}_t\}$ should be close to the observed time series $\{Y_t\}$, i.e., the *residuals*

$$\hat{Z}_t = Y_t - \hat{Y}_t$$

is small. More importantly, $\{\hat{Z}_t\}$ should be similar to the *white noise* sequence $\{Z_t\}$.

Check the goodness of fit of the model by studying the residual

$$\hat{Z}_t = Y_t - \hat{Y}_t$$

Steps

1. Time series plot of $\hat{Z}_t$ (should look like white noises)
2. Plot the ACF of $\hat{Z}_t$, $\hat{r}_Z(j)$.
Good fit if most $\hat{r}_Z(j)$ are within the C.I. $\left(-\frac{2}{\sqrt{n}}, \frac{2}{\sqrt{n}}\right)$, $j \neq 0$.
3. Apply the Ljung-Box test.

Residual Analysis

The Ljung-Box test could be applied to test the goodness of fit. The hypothesis are

H_0 : the series of residuals exhibits no autocorrelation for a fixed number of lag h .

H_1 : some autocorrelation coefficient $\rho_Z(k), k = 1, 2, \dots, h$ are nonzero.

The test statistic is given by

$$Q(h) = n(n+2) \sum_{j=1}^h \frac{\hat{r}_Z^2(j)}{n-j}.$$

- A common choice of h is between 10 to 30
- $Q(h) \rightarrow \chi^2(h-p-q)$. If $Q(h)$ is bigger than the 95% percentile of a $\chi^2(h-p-q)$ distribution, we reject the null hypothesis.
- i.e., good fit if the null hypothesis is not rejected.

Residual Analysis

Example ARMA(1,1): $Y_t = \phi Y_{t-1} + Z_t + \theta Z_{t-1}$.

```
set.seed(4005)
x=arima.sim(1000,model=list(ar=c(0.2), ma=c(0.6)))
fit=arima(x,order=c(1,0,1),include.mean=F)
par(mfrow=c(2,1))
ts.plot(fit$ res)
n=length(fit$ res)
r.z=as.numeric(acf(fit$ res,12)$ acf) ## h=12
Q=n*(n+2)*sum((r.z[-1]^2)/(n-(1:12)))
Q>qchisq(0.95,12-1-1) ## FALSE: not reject  $H_0$ 
```

Alternatively, we can use the built-in function for time-series diagnostics:

```
tsdiag(fit)
```

Four Stages -

1. Pre-processing

- Remove trend and seasonal effect
- Check ACF of the residual to see if it is stationary. Otherwise, take differences to make it stationary

2. Model Estimation (estimating coefficient)

- Draw ACF/PACF of the filtered data and get an initial ARMA model
- MM, YW, LSE, CLS, MLE,

3. Model Identification

- Repeat step 2 for various ARMA models, choose an ARMA model by FPE/AIC/BIC

4. Model Checking (diagnostic)

- Residual analysis (ts.plot/acf of $\{\hat{Z}_t\}$)
- Ljung-Box test